

PS-020

Local SRv6 Path Steering and Measured Failure Recovery

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Problem Description

Hop-by-Hop Routing Inefficiencies: Traditional IPv6 networks route traffic dynamically at every transit node, stripping path control and traffic engineering capabilities from the network edge.

High Convergence Delays: Upon a link failure, standard Interior Gateway Protocols (IGPs) require substantial time to detect the fault, flood topology updates, and recalculate routing tables network-wide.

Packet Loss and Latency: The standard protocol recalculation window causes severe in-flight packet drops, disrupting the ultra-reliable low-latency communications (URLLC) required for mission-critical 6G infrastructures.

Background

Evolution of IPv6 Networks: Modern high-throughput and ultra-reliable low-latency communication (URLLC) environments demand strict quality-of-service and deterministic routing controls that conventional IP infrastructure cannot natively provide.

Limitations of Traditional IGPs: Conventional Interior Gateway Protocols (OSPF/IS-IS) rely on distributed hop-by-hop forwarding decisions. During a link failure, they require significant time for LSA flooding, topology updates, and full Shortest Path First (SPF) recalculations.

Impact on Mission-Critical Traffic: The resulting protocol convergence delay leads to high packet loss, out-of-order delivery, and temporary routing loops during the critical failover window.

Shift to Edge-Controlled Steering: Segment Routing (SRv6) shifts path selection intelligence entirely to the ingress edge node. By encoding the forwarding path as an explicit list of Segment Identifiers (SIDs) in the packet header (RFC 8754), intermediate routers remain stateless while enabling deterministic, real-time path manipulation.

Literature

- **The Industry Benchmark (TI-LFA):** Traditional Topology-Independent Loop-Free Alternate (TI-LFA) relies on localized node-level repair and complex, distributed IGP calculations to achieve fast reroute.
- **Proactive Path Steering (Our Approach):** Shifts failover logic entirely away from core routers. Uses a localized, script-driven mechanism to monitor link-state events strictly from the ingress point.
- **The Optimization Impact:** Directly rewrites the active SID list at the source node upon a trigger event, providing sub-millisecond path steering and bypassing standard IGP convergence overhead entirely.

RFC Review

- **RFC 8402 (Segment Routing Architecture):** Establishes the source-routing paradigm, eliminating per-flow state across core transit routers by maintaining state exclusively at the ingress edge.
- **RFC 8754 (IPv6 Segment Routing Header):** Defines the Type 4 Routing Extension Header carrying the ordered list of 128-bit SIDs. Introduces the Segments Left pointer and specifies ingress encapsulation (T.Encaps).
- **RFC 8986 (SRv6 Network Programming):** Standardizes endpoint behaviors (e.g., End for transit waypoints, End.X for L3 cross-connects). Enables programmable execution where a SID's Locator:Function:Argument structure dictates localized policy execution.

Proposed Solution

- **Edge-Centric Intelligence (Stateless Core):** Migrated all path-computation logic away from core transit routers strictly to the ingress edge (R1). Transit nodes are relieved of complex routing state calculations.
- **Proactive Socket Monitoring (T=0 Detection):** Replaced slow IGP keepalive timers with a custom Python raw-socket listener, achieving instantaneous physical link-state detection directly at the kernel level.
- **Dynamic SRH Rewriting (Sub-ms Recovery):** Programmed a localized, sub-millisecond swap of the active Segment Routing Header (SRH) upon failure. This immediately steers traffic to the backup path, entirely bypassing protocol convergence overhead.

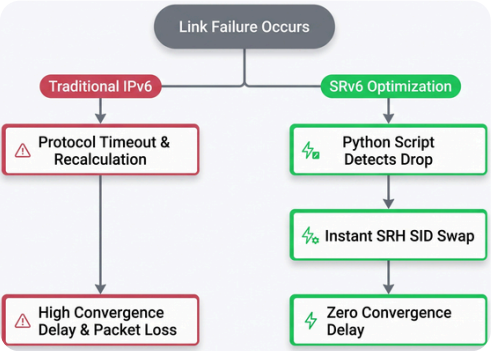
Optimization Proposed

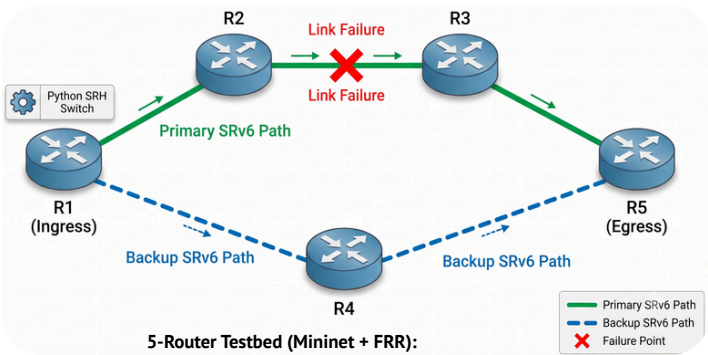
Metric	Traditional IPv6 (Before)	Edge-Steered SRv6 (After)
Failure Detection	Protocol Dead Timers	Real-Time Socket Monitor
Recovery Action	Network-Wide Recalculation	Ingress SRH Rewrite
Convergence Delay	High (Seconds)	Near-Zero (Sub-millisecond)
Packet Loss	Severe	Minimal
Core Network State	Heavy Routing Tables	Stateless Transit

Key Takeaway:

Achieves deterministic, sub-millisecond failover critical for mission-critical traffic by eliminating standard protocol convergence overhead.

Solution Architecture & Design

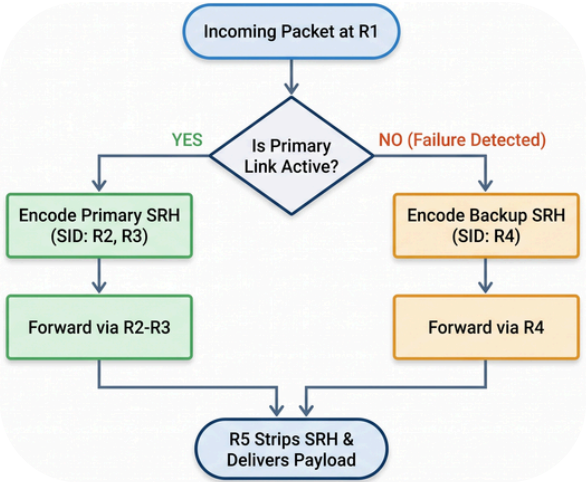




5-Router Testbed (Mininet + FRR):
Simulates an enterprise IPv6 core network with explicit SID routes.

- **Edge Encapsulation (RFC 8754):** Ingress router R1 encapsulates incoming packets with an explicit SRH SID list. Core transit routers (R2, R3, R4) forward packets statelessly without complex routing decisions.
- **Python Backup-Policy Switch:** A lightweight monitoring script runs on R1 to track primary link status (R2–R3).
- **Sub-Millisecond Rerouting:** Upon link failure, the script immediately rewrites incoming packet headers with the backup SID list, steering traffic through R4 to R5 without waiting for IGP convergence.

Data Flow Diagram



Edge-Steering Logic Flow

Methodology

The project utilizes an iterative prototyping methodology within a virtualized Linux sandbox. The team will sequentially establish a standard IPv6 routing baseline, overlay the SRv6 deterministic paths, and integrate the custom Python automation to capture comparative performance metrics.

Deployment Timeline

Phase 1: Blueprint & Infrastructure **(Oct 10-13)**: Repository configuration **(A3-PS020-TC478)**, architecture planning, and structural blueprint submission.

Phase 2: Virtual Testbed & Baseline **(Oct 14-15)**: Mininet topology deployment and ordinary FRRouting (FRR) IPv6 baseline configuration.

Phase 3: SRv6 & Automation Logic **(Oct 16-18)**: Encoding explicit Primary and Backup paths, followed by the deployment of the Python failure-detection script on the ingress router.

Phase 4: Validation & Delivery **(Oct 19-20)**: Controlled failure injection between transit nodes, quantitative metric capture (repair delay, packet loss), and final mentor demonstration.

Bibliography I

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Thank You for Your Attention!